Consistency of Force Prediction in Single Execution Figure 5 displays the alignment between ForceFlow's predicted contact force curve (red) and the real measured force (blue) across six representative tasks. In short-contact tasks (Stamp/Plug/Press), during the contact establishment phase, the rising trend of the predicted force aligns highly with the real contact force. During the contact release phase (task completion or leaving the surface), the predicted force declines synchronously with the real force. This suggests that the model can infer the impending contact state via force history, vision, and proprioception. In continuous tasks (Clean Whiteboard/Clean Vase), the predicted force smoothly adjusts following surface changes, without exhibiting significant high-frequency oscillation or jitter. This indicates that the model is not simply regressing to a mean force value but has learned state-based continuous force regulation.

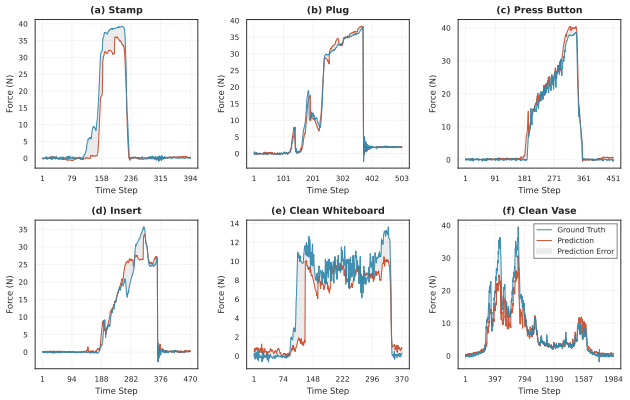

Figure 5. Comparison of Predicted and Measured Forces. Alignment between the predicted (red) and measured ground truth (blue) contact forces. Shaded areas represent prediction error, demonstrating high temporal fidelity during interaction.

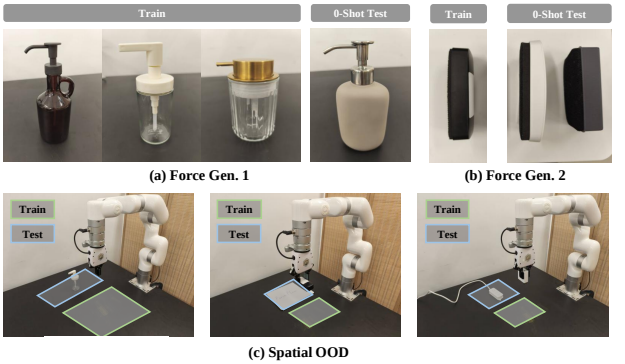

Figure 6. **OOD Evaluation Setup.** (a)-(b) **Force-side Generalization:** Unseen objects (e.g., distinct bottles, different erasers) used for zero-shot testing. (c) **Vision-side Generalization:** The testing workspace (marked "Test") is spatially disjoint from the training distribution (marked "Train"), requiring V2F for localization.

Table 3. **Zero-Shot Force Generalization (SR %).**

Method	Press	Clean WB	Clean Vase
$\pi_{0.5}$	0%	0%	0%
ACT	0%	0%	0%
Diffusion Policy	0%	0%	0%
ForceVLA	40%	90%	0%
ForceFlow (Ours)	**80%**	**100%**	**60%**

As shown in Table 3, vision-centric baselines fail entirely when tool friction or object elasticity changes, as they cannot perceive high-frequency physical feedback. In contrast, ForceFlow maintains high success rates by adaptively adjusting contact forces through its temporal force perception and active prediction mechanisms. By encoding F_{hist}, the model perceives contact states and achieves proactive